

DevOps Learning Roadmap

Based on: "How I Would Learn DevOps from Scratch If I Could Start Over" — Cloud Jam | Improved & Updated Roadmap

■ What is DevOps?

DevOps bridges **Developers** and **Operations** teams to deliver software faster, with fewer bugs and higher quality. Everything in DevOps revolves around three pillars:

■ ■ Infrastructure	The underlying systems & hardware needed for software delivery (servers, databases, networks).
■ ■ Automation	Using tools & scripts (Terraform, Ansible) to replace manual, repetitive work and speed up delivery.
■ Monitoring	Tracking application health, detecting issues, and fixing them before customers are affected.

■ ■ The Learning Roadmap — Step by Step

01 Linux

- Learn the Linux file system, permissions, and text editors
- Master essential shell commands used daily by DevOps engineers
- Practice basic shell scripting for automation
- Tip: Replace Windows with Linux on your machine to learn faster

02 Git

- Start versioning your work from day one — every project
- Git: clone, commit, push, pull, branch, merge, pull requests
- Use GitHub / GitLab for version control and collaboration
- Original roadmap placed Git too late; it belongs here

03 Networking

- DNS — A records, CNAME, TXT records
- TCP/IP protocols and common ports (SSH: 22, MySQL: 3306, RDP: 3389)
- Gateways, NAT, and routing tables
- Subnetting / CIDR notation for IP distribution

04 Python Scripting

- Python basics: data types, functions, file handling, libraries
- Write scripts to automate repetitive DevOps tasks
- Focus on automation use cases, not software development
- Tip: certs are useful but real projects prove more to recruiters

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Cloud (AWS / Azure / GCP)

- Start with AWS Cloud Practitioner Essentials (foundational cert)
- Progress to Solutions Architect Associate or Developer Associate
- Focus on ~20-30 core services: EC2, S3, RDS, VPC, Lambda, ECS, EKS, SNS, SQS, Secrets Manager
- Learn to deploy servers, databases, and load balancers manually first

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Infrastructure as Code

- Terraform: providers, resources, variables, outputs, provisioners
- Automate cloud infra provisioning instead of clicking through consoles
- Ansible: configure multiple servers simultaneously (configuration management)
- Hands-on projects matter more than the cert badge alone

07

CI/CD Pipelines

- Understand Continuous Integration & Continuous Deployment principles
- Pipeline stages: Build → Test → Deploy
- Tools: GitLab CI, Jenkins, CircleCI, GitHub Actions
- Build a project: auto-build/test on every push, deploy safely

08

Containerization

- Docker: images, containers, Dockerfile, essential Docker commands
- Kubernetes: architecture, deployments, namespaces, pod management
- Understand containers vs VMs and why every modern app uses them
- Start with basics; deepen knowledge on the job

09

Monitoring & Observability

- Prometheus + Grafana: metrics collection and dashboards
- Centralized logging: ELK stack (Elasticsearch, Logstash, Kibana) or Grafana Loki
- Distributed tracing with OpenTelemetry — now industry standard
- Original roadmap mentioned monitoring in theory but never taught it

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DevSecOps

- IAM least-privilege: never give more permissions than needed
- Secrets management: HashiCorp Vault or AWS Secrets Manager
- Vulnerability scanning in CI/CD pipelines (Trivy, Snyk)
- Security is no longer optional — expected from DevOps engineers in 2025+

■ Beyond Technical Skills

■ Hands-on Projects

Build real projects for every tool you learn. Recruiters value demonstrated skills over certifications alone.

■ Learn in Public

Post on LinkedIn, write blogs, make videos. Showcasing your learning attracts recruiters organically.

■ Soft Skills	DevOps engineers collaborate with dev, QA, and management teams daily. Communication and teamwork are non-negotiable.
■ Continuous Learning	DevOps evolves constantly. Attend events, join communities, and never stop experimenting with new tools.

Original source: *"How I Would Learn DevOps from Scratch If I Could Start Over"* — Cloud Jam | youtube.com/watch?v=EAXdnPWUCcc
| Roadmap improved with Git reordering, Monitoring/Observability & DevSecOps additions

DevSecOps Learning Roadmap

A step-by-step path to add security to the DevOps skills you already have

DevSecOps is not a replacement for DevOps — it is the same workflow with security automated into every stage. Step 10 of the DevOps roadmap above introduced it; the roadmap below expands it into its own learning path. Tip: you don't need every tool — the bold tool in each card is the easiest free place to start, and **Trivy** alone covers steps 3, 5 and 6.

■ The DevSecOps Roadmap — Step by Step

01

Security Foundations

- Learn the OWASP Top 10 — the most common web vulnerabilities
- Understand the CIA triad: Confidentiality, Integrity, Availability
- Internalise the “shift-left” mindset: catch issues early
- Know the difference between SAST, DAST, SCA before using tools

02

Secure Coding & SAST

- Recognise risky patterns: SQL injection, XSS, hard-coded secrets
- Run static analysis on your own source code on every commit
- Tools: Semgrep (start here), SonarQube, CodeQL
- Practice: fix a flagged vulnerable query into a safe one

03

Dependency & Supply-Chain (SCA)

- Scan open-source libraries for known CVEs — most breaches start here
- Generate an SBOM (Software Bill of Materials) for your app
- Tools: Trivy, Snyk, OWASP Dependency-Check
- Decide: upgrade, patch, or accept each finding

04

Secrets Management

- Stop hard-coding passwords, API keys, and tokens in the repo
- Store secrets in a vault and inject them at runtime
- Block accidental commits with a secret scanner
- Tools: HashiCorp Vault, AWS Secrets Manager, GitLeaks

05

Container & Image Security

- Scan Docker images layer-by-layer before pushing them
- Harden images: minimal/distroless base, never run as root
- Tools: Trivy, Grype, Clair
- Tie scanning into the build so bad images never deploy

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Infrastructure-as-Code Security

- Treat Terraform / Kubernetes manifests as code to be scanned
- Catch misconfigs: public buckets, unencrypted DBs, open ports
- Tools: Checkov, tfsec, Terrascan
- Fail the plan stage on critical misconfigurations

07

Security in the CI/CD Pipeline

- Wire the scanners above into your existing pipeline
- Set fail thresholds (e.g. block on HIGH & CRITICAL only)
- Triage false positives so the pipeline stays trusted
- Sign artifacts & images for provenance (Cosign / Sigstore)

08

Runtime & Cloud Security

- DAST: attack the running app in a test env (OWASP ZAP)
- IAM least-privilege — never grant more access than needed
- Cloud posture (CSPM) to find risky cloud settings
- Runtime threat detection: Falco for containers/Kubernetes

09

Policy as Code & Compliance

- Write security rules as code that the pipeline enforces
- Examples: “no root containers”, “all data encrypted”
- Tools: OPA (Open Policy Agent), Kyverno
- Produce audit trails for standards (SOC 2, ISO 27001, PCI)

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Practice & Stay Current

- Build an end-to-end secure pipeline as a portfolio project
- Follow CVE feeds and security advisories regularly
- Optional cert: Certified Kubernetes Security Specialist (CKS)
- Security evolves fast — keep experimenting with new tools

■ Putting It Together — One Secure Pipeline

Commit → Gitleaks (secrets) → Semgrep (SAST) → Trivy (dependencies) → build → Trivy (image) → Checkov (IaC) → OPA policy gate → **deploy** → ZAP + Falco (runtime). Every arrow is automated; any step can fail the build so insecure code never ships.

Added section: DevSecOps Learning Roadmap — appended to the DevOps Learning Roadmap.